Savitribai Phule Pune University

S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination

Lab Course CS-253-MJ-P (Lab Course based on CS-251 MJ-T & CS-252-MJ-T)

Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Implement a Binary search tree (BST) library (btree.h) with operations – create, search, insert, inorder, preorder and postorder. Write a menu driven program that performs the above operations.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and stores it as an adjacency matrix. Display the adjacency matrix.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Entities and their Relationships for Project-Employee database.
Project (pno integer, pname char (30), ptype char (20), duration integer)
Employee (eno integer, ename char (20), qualification char (15), joining_date date)

The Relationship between Project and Employee is **many to many** with descriptive attribute start_date date, no_of_hours_worked integer.
Constraints: Primary Key, duration should be greater than zero, pname should not be null.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to accept eno as input parameter and count number of projects on which that employee is working.

OR

1) Write a trigger before inserting into a project table to check duration should be always greater than zero. Display appropriate message

2) Write a procedure to accept three numbers from user and display addition, subtraction and multiplication of three numbers.

[5 marks]

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to find the minimum and maximum values in a Binary Search Tree

OR

Q1) Write a C program that accepts the vertices and edges of a graph. Create and display adjacency list also print indegree, outdegree and total degree of all vertex of graph.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Entities and their Relationships
Person (pno integer, pname varchar (20), birthdate date, income money)
Area (aname varchar (20), area_type varchar (5))
An area can have **one or more persons** living in it, but a person belongs to exactly one area.

Constraints: Primary Key,area_type can be either **'urban' or 'rural'.**

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to print total number of persons of a particular area. Accept area name as input parameter.

OR

1) Write a stored function to print sum of income of person living in " " area type. (Accept area type as input parameter). Display appropriate message for invalid area type.

2) Write a procedure to accept two numbers from user and display division of two numbers. Use raise to display error messages for division by zero error. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency matrix**. Implement function to traverse the graph using Breadth First Search (BFS) traversal.

OR

Q1) Write a C program to find the height of the tree and check whether given tree is balanced or not.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Entities and their Relationship.

Student (s_no integer, s_name char (20), address char (25), class char (10))

Teacher (t_no integer,t_name char (10), qualification char (10),experience integer)

Relationship between Student and Teacher is **Many to Many** with descriptive attribute subject and marks_scored.

Constraints: Primary Key,s_name,t_name should not be null,marks_scored> 0

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a cursor which will accept student number from the user and display student name along with teacher name who taught ‘RDBMS’ subject.

OR

1) Write a trigger before insert the record of student. If student number is less than or equal to zero give message “Invalid Number”.
2) Write a procedure accept two numbers from user and display minimum and maximum from two numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to find the height of the tree and check whether given tree is balanced or not.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency list.** Implement function to traverse the graph using Depth First Search (DFS) traversal.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Entities and their Relationships .

Movies (m_name varchar (25), release_year integer, budget money)

Actor (a_name char (30), role char (30), charges money, a_address varchar(30))

Producer (producer_id integer, name char (30), p_address varchar (30))

The relationships are as follows:

Each actor has acted in one or more movies. Each producer has produced many movies and each movie can be produced by more than one producers. Each movie has one or more actors acting in it, in different roles.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a cursor to pass actor name as input parameter and return total number of movies in which given actor is acting.

OR

1) Write a trigger before inserting into movie table to check budget. Budget should be minimum 50 lakh. Display appropriate message.

2) Write a procedure to accept a number from user and check the number is positive, negative or zero. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program which uses Binary search tree library and display the following

i. Node value at each level
ii. Count of node at each level
iii. Total levels in the tree.

OR

Q1) Write a C program for the Implementation of Prim's Minimum spanning tree algorithm.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Entities and their Relationships for Student-Competition
Student (sreg_no int ,s_name varchar(20), s_class char(10))

Competition (c_no int ,c_name varchar(20), c_type char(10))

Relationship between Student and Competition is many to many with descriptive attributes rank and year.

Constraints: Primary Key, c_type should not be null, c_type can be 'sport' or 'academic'.

Q2) Using above Database Solve the following **[10 Marks]**

1. write a function using cursor to accept year and class to display student name, competition name and rank.

OR

1. Define a trigger before updating a competition table. Raise a notice and display message "competition record is being updated".

2. Write a procedure to accept three numbers from user and find maximum and minimum from it. **[5 marks]**

Q 3) Viva **[5 marks]**

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Lab Course CS-253-MJ-P (Lab Course based on CS-251 MJ-T & CS-252-MJ-T)

Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the Implementation of Kruskal's Minimum spanning tree algorithm.

OR

Q1) Write a C program which uses Binary search tree library and implements following function with recursion:

int copy(T) – create another BST which is exact copy of BST which is passed as parameter.

int compare(T1, T2) – compares two binary search trees and returns 1 if they are equal and 0 otherwise.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships

BUS (bus_no int , capacity int , depot_name varchar(20))

ROUTE (route_no int, source char(20), destination char(20),no_of_stations int)

DRIVER (driver_no int , driver_name char(20), license_no int, address char(20), d_ageint , salary float)

The relationships are as: BUS_ROUTE: M-1

BUS_DRIVER: M-M with descriptive attributes Date of duty allotted and Shift itcan be 1 (Morning) or 2 (Evening**).**

Constraints: 1. License_no is unique.

2. Bus capacity is not null

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function to accept the bus number and print driver name allotted to that bus.

OR

1. Write a stored function to accept depot name and display driver details having age more than 50.

2. Write a procedure that accept any number from user and find number is even or odd. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the implementation of Topological sorting.

OR

Q1) Write a C program for the implementation of Dijkstra's shortest path algorithm for finding shortest path from a given source vertex using adjacency cost matrix.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships
Branch (br_id integer, br_name char (30), br_city char (10))
Customer (cno integer, c_name char (20), caddr char (35), city char (20))
Loan_application(lno integer, l_amt_required money, l_amt_approved money, l_date date)
Relationship between Branch, Customer and Loan_application is Ternary. Ternary (br_id integer, cno integer, lno integer) Constraints: Primary Key, l_amt_required should be greater than zero.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function using cursor to count the number of customers of particular branch. (Accept branch name as input parameter).

OR

1. Write a trigger before insert record of customer. If the customer number is less than or equal to zero, then give the appropriate message.

2. Write a procedure to accept value of n and display sum of first n numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Lab Course CS-253-MJ-P (Lab Course based on CS-251 MJ-T & CS-252-MJ-T)

Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a program to sort n randomly generated elements using Heapsort method.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and stores it as an adjacency matrix. Display the adjacency matrix.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships

TRAIN: (train_no int, train_name varchar(20), depart_time time , arrival_time time, source_stn varchar (20),dest_stn varchar (20), no_of_res_bogies int ,bogie_capacity int)

PASSENGER : (passenger_id int, passenger_name varchar(20), address varchar(30), age int ,gender char)

Relationships:

Train _Passenger: M-M relationship named ticket with descriptive attributes as follows

TICKET: (train_no int, passenger_id int, ticket_no int ,bogie_no int, no_of_berths int ,tdate date , ticket_amt decimal(7,2),status char)

Constraints: The status of a berth can be 'W' (waiting) or 'C' (confirmed).

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function to display the ticket details of a train. (Accept train name as input parameter). Raise an exception in case of invalid train name.

OR

1.Write a trigger after insert on passenger to display message "Age above 5 will be charged full fare" if age of passenger is more than 5.

2. Write a procedure that accept two numbers and display addition, subtraction and multiplication of two numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to implement a hash table using **separate chaining** with linked lists with following operation.

a. Insert a key
b. Search a key
c. Display the hash table

OR

Q1) Write a C program to find the height of the tree and check whether given tree is balanced or not.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following database maintained by a school about students and competitions.
STUDENT (sreg_no int , name char(30), class char(10))
COMPETITION (c_no int , name char(20), C_type char(15**))**

The relationship is as follows:

STUDENT-COMPETITION: **M-M** with described attributes rank, year and prize(int).

Q2) Using above Database Solve the following **[10 Marks]**

1. Define a trigger before updating a competition table. Raise a notice and display message "competition record is being updated".

OR

1. Write a function using cursor which accepts a name of student and returns the total no of prizes won by that student in the year 2020.

2. Write a procedure to accept two numbers from user and display division of two numbers use raise to display divided by zero exception messages. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the implementation of Floyd Warshall's algorithm for finding all pairs shortest path using adjacency cost matrix.

OR

Q1) Write a C program to find the minimum and maximum values in a Binary Search Tree.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships
Book(b_no int, b_name varchar (20), pub_name varchar (10), b_price float)
Author (a_no int, a_name varchar (20), qualification varchar (15), address varchar (15))

Relationship between Book and Author is many to many.
Constraints: Primary Key,pub_name should not be null.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function to display the book details written by author . (Accept Author name as input parameter). Raise an exception in case of invalid author name.

OR

1. Write a trigger after insert on book to display message "prize is so high" if book price is more than 1000.
2. Write a procedure to display all even numbers from 1 to 50. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a program which uses binary search tree library and counts the total nodes and total leaf nodes in the tree.

int count(T) – returns the total number of nodes from BST
int countLeaf(T) – returns the total number of leaf nodes from BST

OR

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency list.** Implement function to traverse the graph using Breadth First Search (BFS) traversal.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships

Student (rollno, s_name , class)

Subject (scode , subject_name)

Relationship between Student and Subject is **many to many** with descriptive attribute marks.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function using cursor to calculate total marks of each student and display it.

OR

1. Define a trigger before deleting a student record from student table. Raise a notice and display message "student record is being deleted".

2. Write a procedure to accept value of n and find sum and average of first n number.

[5 marks]

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a program to implement various types of hash functions which are used to place the data in a hash table

a. Division Method
b. Mid Square Method
c. Digit Folding Method

Accept n values from the user and display appropriate message in case of collision for each of the above functions.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an adjacency list. Implement function to traverse the graph using Depth First Search (DFS) traversal.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Project-Employee database maintained by a company which stores the details of the projects assigned to the employees.

Following are the tables:

PROJECT (pno integer, p_name char (30), ptype char(20),duration integer)

EMPLOYEE (eno integer, e_name char (20), qualification char (15), joindate date)

The relationships are as follows:

PROJECT-EMPLOYEE:**M-M** Relationship, with descriptive attributes as start_date (date), no_of_hours_worked (integer).

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a trigger before inserting into a project table to check duration should be always greater than zero. Display appropriate message.

OR

1. Write a stored function to accept project name as input and print the names of employees working on the project. Raise an exception for an invalid project name.

2. Write a procedure to accept value of m and n and display the total count of odd numbers from m to n.

[5 marks]

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to implement a hash table using **separate chaining** with linked lists with following operation.

a. Insert a key
b. Search a key
c. Display the hash table

OR

Q1) Write a C program that accepts the vertices and edges of a graph. Create and display adjacency list also print indegree, outdegree and total degree of all vertex of graph.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database of Bus transport system. Many buses run on one route. Drivers are allotted to the buses shift-wise.

Following are the tables:

BUS (bus_no int , capacity int , depot_name varchar(20))
ROUTE (route_no int, source char(20), destination char(20),no_of_stations int)
DRIVER (driver_no int , driver_name char(20), license_no int, address char(20), d_age int , salary float)

The relationships are as follows:

BUS_ROUTE : **M-1**
BUS_DRIVER : **M-M** with descriptive attributes Date of duty allotted and Shift – it can be 1 (Morning) 0r 2 (Evening).

Constraints: 1. License no is unique.
2. Bus capacity is not null.

Q2) Using above Database Solve the following **[10 Marks]**

1. Define a trigger before insert the record of driver if the age is not between 18 and 50, raise an error message "invalid entry".

OR

1. Write a stored function to accept the bus_no and date and print its allotted drivers. Raise an exception in case of invalid bus number.

2. Write a procedure to accept value of n and display all odd numbers from 1 to n.

[5 marks]

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to maintain a phonebook using Binary Search Tree by name where each node contains contact name and phone number.

OR

Q1) Write a C program to implement a hash table using **open addressing with linear probing**. perform the following operations. (Assume all keys are positive integers)

a. Insert
b. Search
c. Display

Section II: Database Management Systems-II **[15 Marks]**

Consider the following Project-Employee database maintained by a company which stores the details of the projects assigned to the employees.
Following are the tables:
PROJECT (pno integer, p_name char (30), ptype char(20),duration integer)
EMPLOYEE (eno integer, e_name char (20), qualification char (15), joindate date)

The relationships are as follows:

PROJECT-EMPLOYEE: M-M Relationship, with descriptive attributes as start_date (date), no_of_hours_worked (integer).

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a trigger before inserting into a project table to check duration should be always greater than zero. Display appropriate message.

OR

1. Write a stored function to accept project name as input and print the names of employees working on the project. Raise an exception for an invalid project name.

2. Write a procedure to insert the values in project table. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to implement a hash table using **Open Addressing with Quadratic Probing**. Perform the following operations. (Assume all keys are positive integers)

a. Insert
b. Search
c. Display

OR

Q1) Implement a Binary search tree (BST) library (btree.h) with operations – create, search, insert, inorder, preorder and postorder. Write a menu driven program that performs the above operations.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Person–Area database

Person (pnumber integer, pname varchar (20), birthdate date, income money)
Area (aname varchar (20), area-type varchar (5))
The relationships are as follows:

Person-Area: M-1
The attribute 'area_type' can have values either 'urban' or 'rural'.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a cursor to update the income of all people living in 'Urban' area by 10%.

OR

1. Write a trigger before insert the record of person if the person number is negative raise the message "Invalid Number".
2. Write a procedure to display all odd numbers from 1 to 100. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a program to sort n randomly generated elements using Heapsort method.

OR

Q1) Write a C program for the implementation of Dijkstra's shortest path algorithm for finding shortest path from a given source vertex using adjacency cost matrix.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Student –Subject database

Student(rollno integer, name varchar(30), address varchar(50), class varchar(10))

Subject(scode varchar(10), subject_name varchar(20))

Student-Subject are related with **M-M** relationship with attributes **marks_scored.**

Q2) Using above Database Solve the following **[10 Marks]**

1) Define a trigger before deleting a student record from the student table. Raise a notice and display the message "student record is being deleted".

OR

1) Write a stored function using cursors, to calculate total marks of each student and display it.

2) Write a procedure to accept a number and check the number is Positive and Negative or Zero. **[5 marks]**

Q 3) Viva **[5 marks]**

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Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program that accepts the vertices and edges of a graph. Create and display adjacency list also print indegree, outdegree and total degree of all vertex of graph.

OR

Q1) Write a C program to find the minimum and maximum values in a Binary Search Tree.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database maintained by a school about students and competitions.
STUDENT (sreg_no int , name char(30), class char(10))
COMPETITION (c_no int , name char(20), C_type char(15))

The relationship is as follows:

STUDENT-COMPETITION: **M-M** with described attributes rank , year and prize(int).

Q2) Using above Database Solve the following **[10 Marks]**

1) Define a trigger before updating a competition table. Raise a notice and display message "competition record is being updated".

OR

1) Write a function using cursor which accepts a name of student and returns the total no of prizes won by that student in the year 2020.
2) Write a procedure to find sum of first 50 numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to find the height of the tree and check whether given tree is balanced or not.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and stores it as an adjacency matrix. Display the adjacency matrix.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Item-Supplier database

Item(itemno integer, itemname varchar(20))

Supplier(supplier_no integer, supplier_name varchar(20), city varchar(20))

The relationship is as,

Item-Supplier: **M-M** relationship with rate(money) and quantity (integer) as descriptive attributes.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a cursor to display the names of items whose rate is more than 500.

OR

1. Write a trigger before insert or update on rate field. If the rate is less than 50 then raise the appropriate exception.

2. Write a procedure to accept three numbers and find the maximum number from it. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to maintain a phonebook using Binary Search Tree by name where each node contains contact name and phone number.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency list.** Implement function to traverse the graph using Breadth First Search (BFS) traversal.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships for Project-Employee database.

Project (pno integer, pname char (30), ptype char (20), duration integer)
Employee (eno integer, ename char (20), qualification char (15), joining_date date)

The Relationship between **Project and Employee** is **many to many** with descriptive attribute start_date date, no_of_hours_worked integer.
Constraints: Primary Key, duration should be greater than zero, pname should not be null.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to accept eno as input parameter and count number of projects on which that employee is working.

OR

1) Write a trigger before inserting into a project table to check duration should be always greater than zero. Display appropriate message

2) Write a procedure to accept two numbers and find minimum and maximum from two numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency list.** Implement function to traverse the graph using Depth First Search (DFS) traversal.

OR

Q1) Write a C program for the Implementation of Kruskal's Minimum spanning tree algorithm.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Department-employee database.

Department (dno integer, dname varchar(20),city varchar(20))

Employee (eno integer, ename varchar(20), salary money)

Department and Employee are related with a **one to many** relationships.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a function to accept department name and display the maximum salary of an employee in that department.

OR

1) Write a trigger before insert/update on an employee record. Raise exception if salary <0.

2) Write a procedure to accept a number and display multiplication table of a number.

[5 marks]

Q 3) Viva **[5 marks]**

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Lab Course CS-253-MJ-P (Lab Course based on CS-251 MJ-T & CS-252-MJ-T)

Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the implementation of Topological sorting.

OR

Q1) Write a C program to implement a hash table using **open addressing with linear probing**. perform the following operations.(Assume all keys are positive integers)

a. Insert
b. Search
c. Display

Section II: Database Management Systems II **[15 Marks]**

Consider the following database

Doctor(d_no int, d_name varchar(30), specialization varchar(35), phone_no int, charges int)

Hospital(h_no int, h_name varchar(20), city varchar(10))

Doctor and Hospital are related with **many to one** relationship.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a function which will accept the name of the hospital and calculate the average charges of doctors visiting that hospital.

OR

1) Write a trigger before insert/update on Doctor. Raise exception if charges are <0.

2) Write a procedure to insert the values in Doctor table. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to implement a hash table using **Open Addressing with Quadratic Probing**. Perform the following operations. (Assume all keys are positive integers)

a. Insert
b. Search
c. Display

OR

Q1) Write a C program that accepts the vertices and edges of a graph and stores it as an adjacency matrix. Display the adjacency matrix.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Entities and their Relationships for Project-Employee database.

Project (pno integer, pname char (30), ptype char (20), duration integer)
Employee (eno integer, ename char (20), qualification char (15), joining_date date)

The Relationship between **Project and Employee** is **many to many** with descriptive attribute start_date date, no_of_hours_worked integer.
Constraints: Primary Key, duration should be greater than zero, pname should not be null.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to accept eno as input parameter and count number of projects on which that employee is working.

OR

1) Write a trigger before inserting into a project table to check duration should be always greater than zero. Display appropriate message

2) Write a procedure to display first 20 odd numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to find the height of the tree and check whether given tree is balanced or not.

OR

Q1) Write a C program for the implementation of Topological sorting.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database

Car (c_no int, owner varchar(20), model varchr(10), color varchar(10)

Driver (driver_no int, driver_name varchar(20), license_no int, d_age int, salary float)

Car and Driver are related with **many to many** relationship

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a cursor which accepts the driver name and prints the details of all cars that this driver has driven, if the driver name is invalid, print an appropriate message.

OR

1) Write a trigger before insert/update on Driver. Raise exception if driver age is < 21.

2) Write a procedure to find sum of first 100 numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the implementation of Floyd Warshall's algorithm for finding all pairs shortest path using adjacency cost matrix.

OR

Q1) Write a C program to maintain a phonebook using Binary Search Tree by name where each node contains contact name and phone number.

Section II: Database Management Systems II **[15 Marks]**

Consider the following Department-employee database.

Department (dno integer, dname varchar(20),city varchar(20))

Employee (eno integer, ename varchar(20), salary money)

Department and Employee are related with a **one to many** relationship

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a function to accept department name as input and display employee name along with salary of that department.

OR

1) Write a trigger after insert on an employee record. Display appropriate message when the record is inserted.

2) Write a stored procedure to accept value of n and display first n even numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the implementation of Dijkstra's shortest path algorithm for finding shortest path from a given source vertex using adjacency cost matrix.

OR

Q1) Implement a Binary search tree (BST) library (btree.h) with operations – create, search, insert, inorder, preorder and postorder. Write a menu driven program that performs the above operations.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following database

Customer (cno integer, cname varchar(20), city varchar(20))

Account (a_no int, a_type varchar(10), opening_date date, balance money)

Customer and Account are related with **one to many** relationship

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a function using cursor which accepts city name as input and prints the details of all customers in that city.

OR

1. Write a trigger which does not allow deletion of accounts of _________type.

2.Write a procedure to display all customers from 'Pune' city. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency matrix**. Implement function to traverse the graph using Depth First Search (DFS) traversal.

OR

Q1) Write a C program to find the minimum and maximum values in a Binary Search Tree.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database :

Item (itemno integer, Itemname varchar(20), quantity integer)

Supplier(supplierno integer ,Supplier name varchar(20),city varchar(20))

Item and supplier are related with many to many relationship. Rate is descriptive attribute.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function using cursors, to accept Item name from the user and display the Rate and Supplier Name for that Item.

OR

2) Write a trigger before update on rate field. If the difference in the old rate and new rate is more than Rs 2000, raise an exception and display the corresponding message.

3) Write a procedure to accept and display addition, subtraction, multiplication and division of two numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the Implementation of Prim's Minimum spanning tree algorithm.

OR

Q1) Write a C program that accepts the vertices and edges of a graph and store it as an **adjacency matrix**. Implement function to traverse the graph using Breadth First Search (BFS) traversal.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database :

Student (rollno integer, name varchar(30),class varchar(10))

Subject(Scode varchar(10),subject name varchar(20))

Student and subject are related with **M-M** relationship with attributes marks_scored.

Q2) Using above Database Solve the following **[10 Marks]**

1. Write a stored function using cursors, to accept class from the user and display the details of the students of that class.

OR

1. Write a trigger before insert/update the marks_scored. Raise exception if Marks are negative.

2. Write a procedure to accept and display addition, subtraction and division of two numbers. Handle division by zero error for division operation. (Use raise). **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program to implement a hash table using **Open Addressing with Quadratic Probing**. Perform the following operations. (Assume all keys are positive integers)

a. Insert
b. Delete
c. Display

OR

Q1) Write a C program for the Implementation of Prim's Minimum spanning tree algorithm.

Section II: Database Management Systems-II **[15 Marks]**

Consider the following database :

Company (Name varchar(30),address (50),city varchar(20), share_value money)

Person (pname varchar(30),pcity varchar (20))

Company and **Person** are related with **M to M** relationship with descriptive attribute No_of_shares

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to update the share_values by 20% for Person "_____".

OR

1) Write a trigger before deleting company record. Display appropriate message to the user.

2) Write a procedure to insert the values in person table. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a program to sort n randomly generated elements using Heapsort method.

OR

Q1) Write a C program for the Implementation of Kruskal's Minimum spanning tree algorithm.

Section II: Database Management Systems II **[15 Marks]**

Consider the following database

Person (pno int, pname varchar (20), birthdate date, income money)

Area (aid int,aname varchar (20), area_type varchar (5))

The **person and area** related to **many to one** relationship. The attribute 'area_type' can have values either 'urban' or 'rural'.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a cursor to display the names of persons living in urban area.

OR

1) Write a trigger before deleting a person's record from the person's table. Raise a notice and display the message "person record is being deleted".
2) Write a procedure to accept three numbers from user and display maximum and minimum of three numbers. **[5 marks]**

Q 3) Viva **[5 marks]**

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Duration: 3 Hours **Maximum Marks: 35**

Section I: Data Structures-II **[15 Marks]**

Q1) Write a C program for the Implementation of Kruskal's Minimum spanning tree algorithm.

OR

Q1) Write a C program to implement a hash table using **open addressing with linear probing**. perform the following operations. (Assume all keys are positive integers)

a.Insert
b.Search
c.Display

Section II: Database Management Systems II **[15 Marks]**

Consider the following database

Student (Roll_No int, Sname varchar (20), Sclass char (10))

Teacher (T_No int, Tname char (20), Experience int)

Student and Teacher are related with **many to many** relationship with the descriptive attribute Subject.

Q2) Using above Database Solve the following **[10 Marks]**

1) Write a stored function to count the number of teachers having experience > 10 years

OR

1) Write a trigger before insert the record of the student in the Student table. If the Roll_No is less than or equal to zero then the trigger gets fired and displays the message "Invalid Roll Number".

2) Write a procedure to accept value of n from user and find sum and average of first n numbers. **[5 marks]**

Q 3) Viva **[5 marks]**